

Global Software

Microsoft Corp

Rating

Outperform

Price Target



660.00 USD (647.00 OLD)



Mark L. Moerdler, Ph.D.
+1 917 344 8506
mark.moerdler@bernsteinsg.com



Firoz Valliji, CFA
+1 917 344 8316
firoz.valliji@bernsteinsg.com



Shelly Tang, CFA
+1 917 344 8342
shelly.tang@bernsteinsg.com

Microsoft: Are they overbuilding capacity?

One of the parts of the Microsoft bear case has been that it is massively spending cash to build AI capacity that will far outstrip demand. This past quarter's results incrementally assuaged that fear, but we continue to receive questions surrounding this concern. Today's note shows that 1) MSFT has added datacenter SQ FT slower than Cloud revenue; 2) MSFT has increased their future lease obligations, but it is spread over so many years that capacity will continue to grow at a benign rate; and 3) while MSFT has purchase obligations (mostly power and cooling but also hardware) on order for the next 12 months they have almost nothing beyond that. Even if AI demand were to stop Microsoft's future datacenter capacity could be shifted to supporting the growth of their existing CPU based Cloud business.

Simply stated Microsoft is not building too fast, but rather taking a surprisingly measured approach given the demand signals they are receiving and their ability to easily pivot facilities to meet demand (CPU or GPU). With capacity coming on-line quickly; hardware generally available as needed (even though some costs have gone up) the setup for the stock is really strong, and therefore we are raising our TP.

Investment Implications

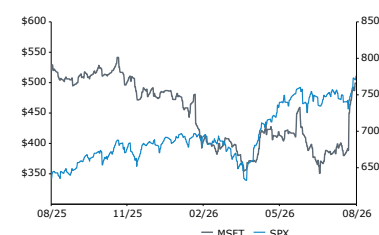
Microsoft (MSFT, TP \$660, Outperform): Microsoft is building flexible capacity and only buying server and related hardware to meet demand. With the future growth rate of datacenter capacity roughly in-line with the historical growth rate of their Cloud business and future lease obligations spread over as long as 20 years MSFT is not overbuilding datacenters. In addition, with no long term obligations to buy meaningful AI capacity Microsoft is procuring equipment to meet demand. Realizing that a meaningful part of the bear case is inaccurate, we strongly reiterate our Outperform rating and raise our target P/E multiple ½ turn to 27x, given what we learned.

Adjusted EPS	F26A	F27E	F28E
MSFT (USD)	17.28	19.96	24.01

Financials	F26A	F27E	F28E	CAGR
Revenues (M)	331,839	392,630	466,335	18.5%
Operating Earnings (M)	155,237	187,110	222,918	19.8%
Net Earnings (M)	128,786	148,728	179,591	18.1%
Revenue/Share	44.52	52.68	62.34	18.3%
Operating Margin (%)	46.8	47.7	47.8	--
Net Income Margin (%)	38.8	37.9	38.5	--
EPS Growth (%)	26.7	15.5	20.3	--

Close Date	7 Aug 2026			
MSFT Close Price (USD)	499.99			
Price Target (USD)	660.00			
Upside/(Downside)	32%			
52-Week Range	553.72/349.20			
SPX	7,757.64			
FYE	Jun			
Div Yield	0.7%			
Market Cap (USD) (B)	3,712.70			
EV (USD) (B)	3,764.67			
Performance	YTD	1M	6M	12M
Absolute (%)	3.4	28.6	24.6	(4.0)
SPX (%)	13.3	3.4	11.9	22.4
Relative (%)	(9.9)	25.2	12.7	(26.4)
Source: Bloomberg, Bernstein estimates and analysis.				

Price Performance, 1YR



Source: Bloomberg, Bernstein estimates and analysis.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our [Global Research Disclosure Website](#).

First Published: 10 Aug 2026 05:29 UTC Completion Date: 10 Aug 2026 05:29 UTC

www.bernsteinresearch.com

VALUATION COMPS TABLE

8/9/2026		Net Cash Per Share		Free Cash Flow (LTM)		Revenue Growth	Valuation Multiples				
Bernstein Coverage	Company	Share Price	\$	% of Share Price	Per Share (\$)	% Yield	NTM (%)	EV / NTM Revenue	P / FE	Price / FCF (LTM)	PEG Ratio
	Adobe	\$265.21	(3.48)	(1.3%)	24.86	9.4%	10.1%	3.9x	10.2x	10.3x	0.8x
	Microsoft	\$499.99	(6.97)	(1.4%)	8.99	1.8%	17.8%	9.6x	25.3x	55.4x	1.8x
	Oracle	\$147.02	(46.51)	(31.6%)	(8.13)	N/A	33.0%	6.2x	18.2x	N/A	3.2x
	MongoDB	\$398.80	28.66	7.2%	7.15	1.8%	18.6%	9.6x	62.3x	54.3x	3.0x
	Salesforce.com	\$192.74	(32.98)	(17.1%)	15.67	8.1%	10.0%	4.0x	13.8x	10.8x	13.3x
	HubSpot	\$210.45	21.13	10.0%	12.64	6.0%	14.0%	2.4x	14.0x	15.9x	0.5x
	SAP	\$205.73	1.09	0.5%	7.39	3.6%	11.5%	5.9x	23.2x	29.3x	1.4x
	Snowflake	\$330.49	0.54	0.2%	3.43	1.0%	28.7%	17.7x	152.6x	97.9x	2.9x
	Quantinuum	\$58.71	2.17	3.7%	(0.85)	N/A	10.4%	529.9x	N/A	N/A	N/A
Workday	\$179.64	2.13	1.2%	11.25	6.3%	11.1%	4.0x	16.1x	14.9x	1.1x	
Other Comp	Autodesk	\$249.08	0.93	0.4%	12.75	5.1%	11.9%	6.2x	19.3x	19.3x	1.3x
	ServiceNow	\$124.88	(3.64)	(2.9%)	4.39	3.5%	20.0%	7.5x	27.4x	28.2x	1.3x
	Amadeus	\$66.46	(7.15)	(10.8%)	4.77	7.2%	8.6%	4.4x	15.6x	13.8x	1.5x
	Datadog	\$233.93	10.14	4.3%	2.93	1.3%	24.4%	16.3x	86.4x	78.5x	6.5x

Source: Bloomberg, Bernstein analysis

Table Of Contents

Portfolio Manager's Summary.....	3
What is the bear case?.....	4
What is Microsoft spending CAPEX and leasing datacenters for?.....	5
How has Microsoft's datacenter capacity grown?.....	6
Understanding current lease liabilities.....	7
Understanding future lease obligations.....	8
Are these obligations non-cancelable and guaranteed?.....	10
What about Microsoft owned datacenters?.....	10
Why are dollar spent growing faster than SQ FT?.....	10
Microsoft's hardware commitments.....	11
What would Microsoft do if supply outstrips demand?.....	11

DETAILS

There is a bear case roiling the valuations across the tech sector that AI datacenter capacity is being massively overbuilt. With Microsoft one of the main concerns has been that the company is over-investing in building capacity (datacenters and hardware) to meet AI demand that may never materialize. The relativity is quite different. While Microsoft is signing long-term leases the capacity is being added over the next 7 years under leases that range from 1 year to 20 years and can be used for growth of their existing Cloud or AI based on demand. Similarly, they are purchasing equipment to meet demand and most of their purchase commitments are for FY27 and range from power system to cooling systems all of which must be ordered in advance; electrical purchasing agreements; infrastructure services and datacenter hardware. In other words, while the numbers may look large given the time horizon and that most of what they are buying is non-AI specific the company has substantial flexibility to accelerate or decelerate adding capacity based on demand.

PORTFOLIO MANAGER'S SUMMARY

Microsoft's stock was trending downward, but that started to reverse with the Q4 results and guidance. Even with the relatively large recent move the stock is still trading at a meaningful discount to its prior valuation multiple range. As will be discussed in this note, the Microsoft bear case is not only breaking, it is based in some ways on a misunderstanding of the story and the numbers. Today's note is going to show that:

1. Microsoft is adding capacity both for AI and non-AI businesses. Most of Microsoft's Cloud revenue ([Exhibit 6](#)) is coming from non-AI Azure and SaaS and those businesses are growing at a healthy rate. Nonetheless, AI while a smaller % of total revenue currently has a much lower margin which means that it still accounts for ~40% of Cloud COGs. Still, Microsoft's investments are not simply for AI, they are mostly for non-AI and that includes not just new capacity but replacing aging equipment every year.
2. Microsoft has been adding datacenter facilities (SQ FT) at a meaningfully lower rate than they are growing their Commercial Cloud business.
3. That Microsoft's future datacenter leases, when modeled on an annual cost of leasing basis, is growing slower than Microsoft's Commercial Cloud, no less Azure.
4. Microsoft's datacenters are designed for flexibility and can be a mix of CPU servers and GPU servers which means that if AI growth slows Microsoft will use their datacenters to meet Cloud demand. Interestingly enough, while investors are focused on AI they seemed to forget that the shift of on-premise servers to IaaS/PaaS is continuing and requiring growth in SQ FT and servers, storage etc.

5. Microsoft's FY27 hardware commitments include power systems, cooling systems, power and datacenter hardware (e.g. servers, networking equipment and storage) and therefore commitments for hardware are likely to be a fraction of the current purchase commitments. Once one considers that Microsoft is also continuing to grow their existing CPU based Cloud business and replacing existing hardware as it ages out, the amount of purchase commitments dedicated to GPUs is likely relatively small.
6. Future purchase commitments beyond FY27 are relatively small, and we believe are predominantly for power and cooling systems and possibly even power.

In other words, Microsoft is not overbuilding datacenters, even considering the substantial jump in CAPEX and additional leases.

Still there will continue to be concern, but we would recommend that investors remember that other than GPU servers everything else that Microsoft is building, leasing or buying can be used for the traditional Commercial Cloud business which has a long runway of growth ahead. In fact, we would argue that if a large Azure customer were to run into issues Microsoft can even pivot AI assets from them to 1) other customers and 2) Copilots and improving SaaS functionality. In other words, with investments not outgrowing historical norms, demand much higher than supply, and most of the investments usable across Commercial Cloud the downside risk here is minimal.

WHAT IS THE BEAR CASE?

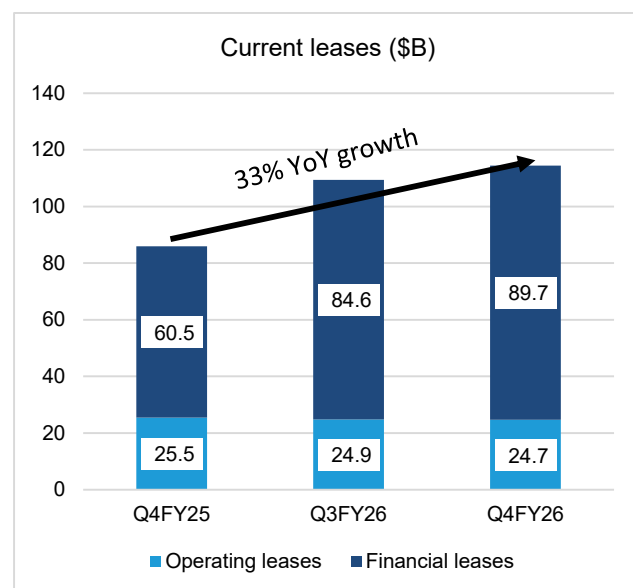
Why has there been concern surrounding Microsoft's lease liabilities and future obligations? Fundamentally investors worry about the return / margin / value Microsoft will capture from all the AI investment (datacenters and hardware). AI is a lower margin business for Microsoft but according to management every AI product's margins are improving, and we believe that will continue. Nonetheless, if the company is overbuilding as the bears are concerned, then a problem could be waiting on the horizon.

The company reports both their current leases, with some details on annual impact, and future lease obligations that have not started. Here are the high level numbers:

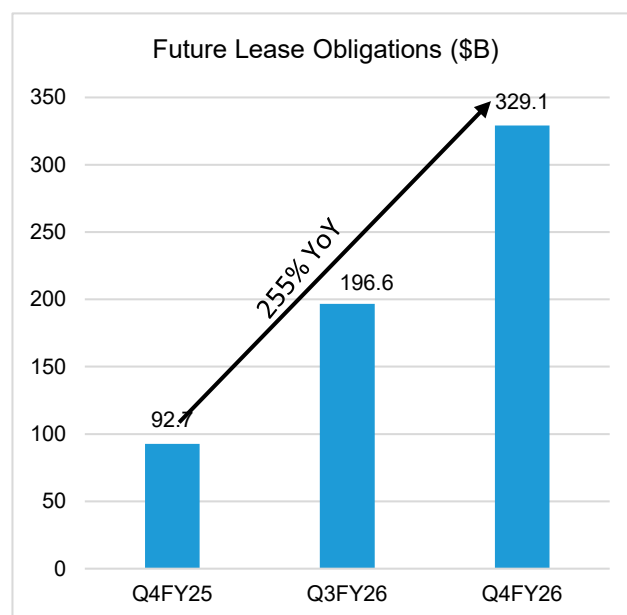
- The total dollar value of Microsoft's current leases has increased 33% YoY this year ([Exhibit 2](#))
- Future lease obligations increased 255% YoY ([Exhibit 3](#)) to reach \$329.1B.
- On the datacenter hardware side, Microsoft does not break down server / hardware obligations, but they give high level information on current year and future purchase commitments ([Exhibit 4](#)) which includes power systems (and probably electrical purchasing commitments), cooling system, non-facilities infrastructure, services, and compute hardware. For FY27, as of the end of FY26, Microsoft had \$169B in total purchase commitments. Beyond FY27 commitments, as of the end of FY26, total \$25.1B.

Those that are worried about the size of Microsoft's commitments add up total current leases + future lease obligations + purchase commitments. But as with so much the devil is in the detail and while the total numbers are large we believe that one should focus on the expense implications on an annual basis and look at those in comparison to the datacenter (CPU + GPU) capacity that Microsoft is able to deliver.

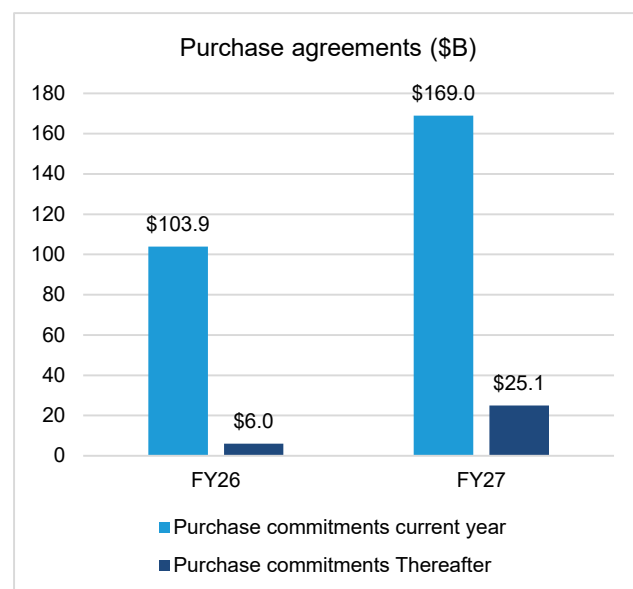
What that calculus misses is that 1) contract duration; 2) future lease obligations go-live over time; and 3) future purchase obligations (beyond 12 months) are small and 4) purchase commitments could be for operating expense items and not CAPEX. In addition, not all the leases will end up happening - one need only read the news to see that legal hurdles are increasing and construction is taking longer than contracted. The remainder of this note will try to place Microsoft's current and future obligations in perspective.

EXHIBIT 2: The lease \$ value of current leases have increased 33% YoY as new datacenters go-live ...

Source: Company reports, Bernstein analysis

EXHIBIT 3: Microsoft's future lease obligations have increased through FY26, reaching \$329.1B

Source: Company reports, Bernstein analysis

EXHIBIT 4: Microsoft is not signing meaningful purchase obligations for power, cooling, hardware and other datacenter items for more than the upcoming year

Source: Company reports, Bernstein analysis

WHAT IS MICROSOFT SPENDING CAPEX AND LEASING DATACENTERS FOR?

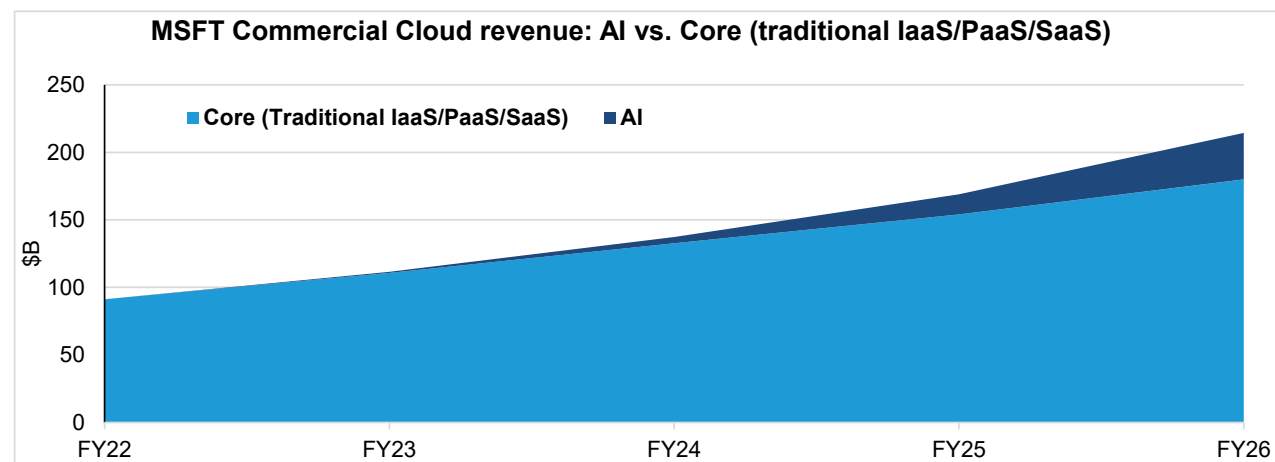
Everyone seems to be focused on how much Microsoft is spending on leases and hardware and assigning this spend to AI. This line of thought ignores the fact that Microsoft's Commercial Cloud revenue, in FY26, delivered \$214.4B growing 9% QoQ in Q4. While Microsoft has not stated how much of FY26 revenue came from AI (Copilots and Azure AI) we know that the AI ARR was \$37B as of Q3 FY26, and we can estimate the commercial Cloud ARR at that time (~\$218B). Leveraging our estimated Microsoft' AI gross margin of ~27% we can estimate that AI could be as much as 40% of COGs exiting Q3 ([Exhibit 7](#)). Note that

management has said that the gross margins for each of Microsoft AI products have been increasing QoQ so over time while AI becomes a larger percentage of revenue it will be much more slowly increasing as a percentage of COGs.

This would argue that ~40% of leases and hardware purchases are for AI but in fact Microsoft has been investing to rebuild datacenters for mixed use and has been replacing servers and other hardware as they age out, so the percentage could be incrementally smaller. One also needs to take into account the fact that SaaS AI is driving CPU revenue and as production enterprise inferencing continues to grow the pull through will increase.

Still, with Azure non-AI growing, SaaS growing and AI driving non-AI workload growth one should not use simply CAPEX, leases, or purchase commitments as a proxy for AI investment.

EXHIBIT 5: AI is ~16-17% of MSFT's Commercial Cloud revenue in FY26



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 6: AI was ~17% of Microsoft's Commercial Cloud ARR as of Q3 FY26

As of Q3 FY26

Microsoft Commercial Cloud ARR (\$B)	218
Microsoft AI ARR (\$B)	37
AI as % of Cloud revenue	17%

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 7: Given the lower margin of MSFT's AI revenue, at this time, we estimate that AI could account for as much as 40% of Commercial Cloud COGs and therefore leases as of Q3 FY26

Microsoft AI ARR (\$B)	37
Estimated AI Gross margin	27%
COGs (\$B)	27

Microsoft Commercial Cloud (\$B)	218
Commercial Cloud gross margins	69%
COGs	68

% of Commercial Cloud COGs from AI	40%
------------------------------------	-----

Source: Company reports, Bernstein analysis and estimates

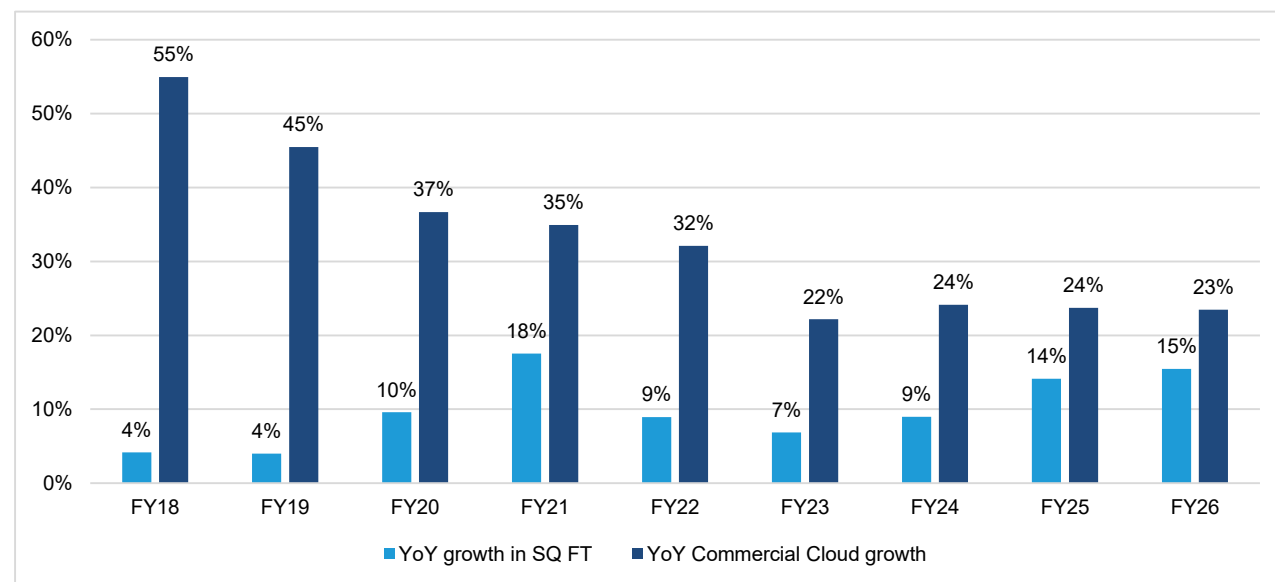
HOW HAS MICROSOFT'S DATACENTER CAPACITY GROWN?

We start with the question of how much has Microsoft been increasing the size of their datacenters. While the company does not disclose how many SQ FT make up their datacenters (or any equivalent number) they supply in the 10Ks the total SQ FT leased and owned. Initially the SQ FT were primarily Microsoft's corporate offices, stores and other similar location but as Microsoft built out Commercial Cloud (Azure, Office 365 etc.) that shifted to mostly datacenters. Since Microsoft is not aggressively adding to the office facilities, stores (which have shrunk over time) and other facilities which make the rate of total growth is a relatively good proxy for the Cloud datacenter build-out.

Looking at the last 8 years of data we find that Microsoft's total SQ FT (owned and lease) has grown slower than Microsoft's Commercial Cloud revenue ([Exhibit 8](#)) even in FY26. Meanwhile, leased SQ FT has grown faster than owned ([Exhibit 9](#)). It is important to note that the percentage of SQ FT owned by Microsoft versus leased is important in this discussion since the cost

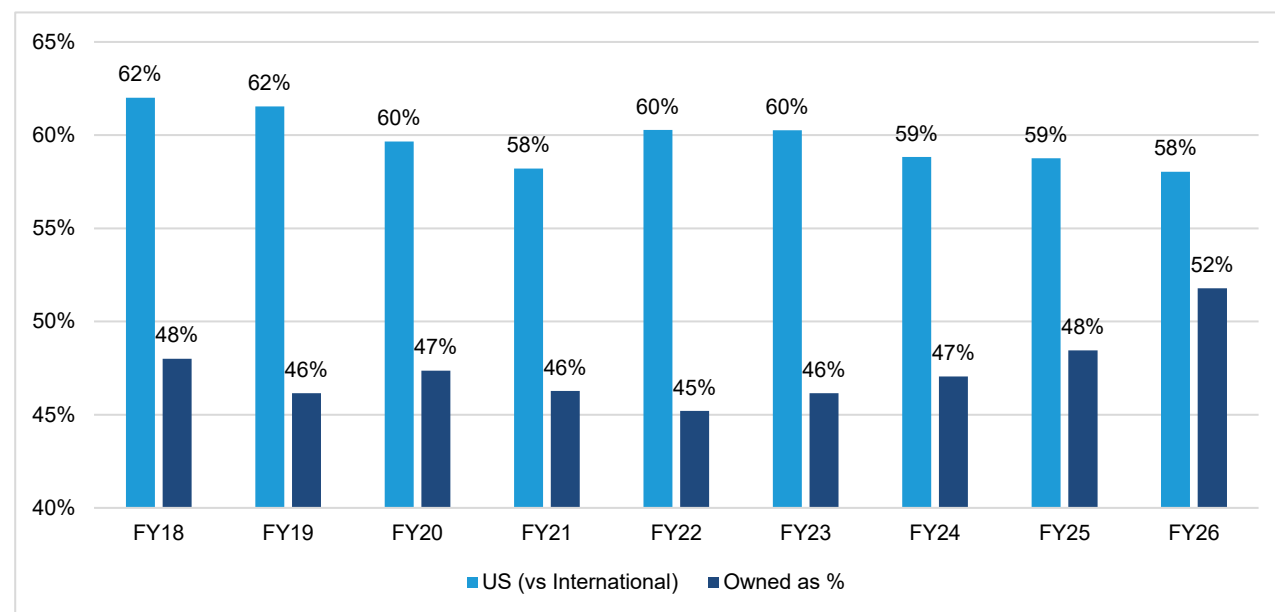
of purchasing / building these facilities are already on the balance sheet and that the cost of maintaining and operating ~ ½ of the datacenters is relatively small, even considering the recent investments in improving datacenters to supply mixed (CPU and GPU) servers.

EXHIBIT 8: Microsoft has been growing total SQ FT slower than Commercial Cloud revenue, even with the AI buildout



Source: Company Reports, Bernstein analysis

EXHIBIT 9: Meanwhile, the % owned has decreased as the % in international locations has increased



Source: Company reports, Bernstein analysis

UNDERSTANDING CURRENT LEASE LIABILITIES

Next we turn to Microsoft's existing (as of the end of FY26) current lease liabilities. To understand what is really going on we believe one needs to understand that Microsoft's current leases, which added up to \$114.4B in Q4FY26, are in fact spread over 13 years ([Exhibit 10](#)). If we break down the leases on an annual expense basis the payments due to existing leases in fact do not

meaningful increase between now and FY31 and based purely on current leases likely decrease over time thereafter.

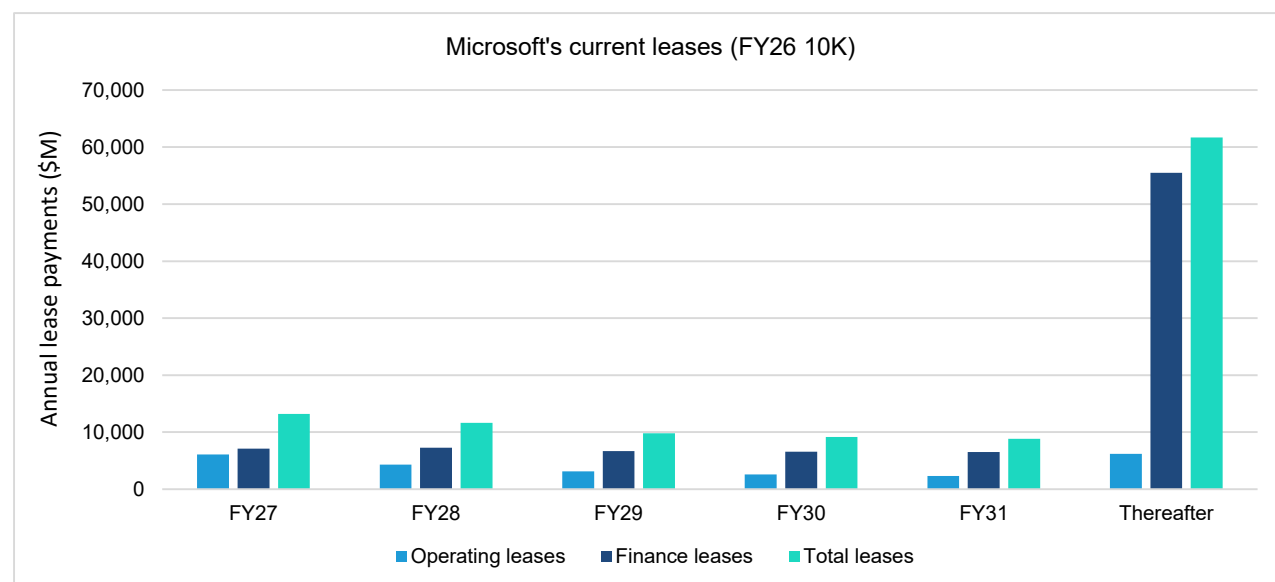
Microsoft has two types of current leases:

- Operating leases which have a weighted average Remaining Lease Term of 6 years
- Finance leases which have a weighted average Remaining Lease Term of 13 years

The original lease terms for each category would have to be at least 1 year longer given this data is based on all the existing leases signed by Microsoft. Fortunately Microsoft supplies the breakdown of how much of the current lease obligation will be realized as an expense over the next 5 years and how much thereafter ([Exhibit 10](#)).

Therefore, while the total obligation of \$114.4B is large the annual payments of \$13.2B in FY27 and decreasing thereafter is not going to be meaningful more than what Microsoft has been paying for leases already (more on this in later in the note).

EXHIBIT 10: Microsoft's current liabilities add up to \$114.4B spread over 13 years



Source: Company reports, Bernstein analysis

As discussed in our [Q4FY26 earnings](#) report, Microsoft has increased the useful life of datacenters from 15 years to 25 years which has an impact on future finance leases and owned datacenters.) While this shifts some future finance leases to operating leases and may extend the useful life of new datacenters (which would amortize the construction investment over a longer period of time) it will not change how much Microsoft will be paying on an annual basis in leases, just how that the obligations are categorized on the balance sheet.

UNDERSTANDING FUTURE LEASE OBLIGATIONS

If the current lease payments are manageable and in fact decreasing each year then where is the concern?

We believe that it is mostly when one considers future lease obligations. Microsoft, in its 10-Qs and 10-Ks, details future lease obligations. In their FY 2026 10K states that:

“As of June 30, 2026, we had additional leases, primarily for datacenters, that had not yet commenced of 329.1 billion, with some arrangements subject to certain contractual conditions being met. These leases will commence between fiscal year 2027 and fiscal year 2033 with lease terms of 1 year to 20 years.”

We believe there are two parts to the concerns:

1. Leases that have been signed but not commenced grew substantial from the end of FY25 to the end of FY26 ([Exhibit 3](#)) reaching \$329.1B or 255% YoY.
2. Current obligations are \$114.4B but the additional, not yet commenced leases are almost 3x the size at \$329.1B

To understand what Microsoft is doing one needs to drill into the details and more specifically two time horizons:

- **Additional datacenters go live over the next 7 years:** These facilities will be turned over to Microsoft, and become a lease obligation between 2027 and 2033.
- **Lease contract obligations can be as long as 20 years:** The lease terms are 1 to 20 years.

To understand what is happening we need to make some assumptions since the company is not supplying details on:

- Do the additional leases spread evenly over the 7-year period of time
- Are all the future leases for datacenters (they could be for other parts of the business or general operations)
- What is the weighted average contract duration;

If we spread the lease go-lives even across the 7-year period of FY27 to FY33 the annual increase in lease obligations would be \$47B. We can then create a sensitivity analysis ([Exhibit 11](#)) showing how much of the \$329.1 converts into lease obligations yearly from FY27 to FY33 and how much thereafter.

If we assume that these are datacenter facilities leases similar to those we discussed in last year's expert call ([Global Software: What is in a Datacenter contract? Key takeaways from discussion with a legal expert](#)) we can assume these are probably mid-teens initial contracts durations with multiple 5-year renewal options. But what is interesting is that Microsoft says these future leases can be anywhere from 1 to 20 years¹ Some are likely specialized capacity agreements and neoclouds at the low end of the range while there is likely only a few really long term commitments.

For the purposes of this discussion we are going to focus on 12-15 year weighted average contract durations for the future lease obligations. It is also possible the average weighted contract duration is shorter as Microsoft has signed agreements with neoclouds and recently with Amazon ([Link](#)).

EXHIBIT 11: Ramping the contract go-lives over 7 years and estimating an average weighted contract duration we can estimate the additional lease obligations

	Average weighted lease duration										
	10	11	12	13	14	15	16	17	18	19	20
2027	4.7	4.3	3.9	3.6	3.4	3.1	2.9	2.8	2.6	2.5	2.4
2028	9.4	8.5	7.8	7.2	6.7	6.3	5.9	5.5	5.2	4.9	4.7
2029	14.1	12.8	11.8	10.8	10.1	9.4	8.8	8.3	7.8	7.4	7.1
2030	18.8	17.1	15.7	14.5	13.4	12.5	11.8	11.1	10.4	9.9	9.4
2031	23.5	21.4	19.6	18.1	16.8	15.7	14.7	13.8	13.1	12.4	11.8
2032	28.2	25.6	23.5	21.7	20.1	18.8	17.6	16.6	15.7	14.8	14.1
2033	32.9	29.9	27.4	25.3	23.5	21.9	20.6	19.4	18.3	17.3	16.5
Thereafter	197.5	209.4	219.4	227.8	235.1	241.3	246.8	251.7	256.0	259.8	263.3

Source: Company reports, Bernstein analysis and estimates

Adding up the existing lease obligations ([Exhibit 10](#)) with the estimated weighted average contract duration for the leases that have not commenced we can estimate the annual payments ([Exhibit 12](#)) and the YoY growth rate that would currently happen if:

¹ This is down from the May 2026 10-K which stated 1 year to 21 years. It is possible a 21 year contract went live or it is possible that the duration changed / was wrong in the filing.

- All the future obligations go-live evenly in FY27-FY33. It is quite likely they do not go-live uniformly, but this is as good an assumption as we can make right now
- The weighted average contract duration is 12–15 years
- None of the existing leases renew for additional periods of time.

We can see a major jump in YoY lease payment growth and therefore likely SQ FT in FY27. While it is likely that the go-live distribution is not uniform, a large YoY jump would mean that Microsoft is gaining a lot more powered SQ FT which would mean that 1) Cloud would accelerate and 2) the demand / supply inequality would mostly disappear.

EXHIBIT 12: Combining the current annual lease obligations (\$B) with our estimated future annual obligations we can model total lease obligations thru FY31 ...

	12 yrs	13 yrs	14 yrs	15 yrs
FY25A	13.9	13.9	13.9	13.9
FY27E	17.1	16.8	16.6	16.3
FY28E	19.5	18.9	18.3	17.9
FY29E	21.6	20.7	19.9	19.2
FY30E	24.9	23.6	22.6	21.7
FY31E	28.8	27.3	26.0	24.9

Source: Bernstein analysis and estimates

EXHIBIT 13: ... and the YoY growth rate in lease expenses as well as an average growth rate which would be relatively in-line with historical norms

	12 yrs	13 yrs	14 yrs	15 yrs
FY27E	23%	21%	19%	17%
FY28E	14%	12%	11%	10%
FY29E	11%	10%	8%	7%
FY30E	15%	14%	14%	13%
FY31E	16%	15%	15%	14%
Average	16%	14%	13%	12%

Source: Bernstein analysis and estimates

ARE THESE OBLIGATIONS NON-CANCELABLE AND GUARANTEED?

If we look at the wording that Microsoft uses in the 10-Qs and 10-K we find two interesting points:

1. Microsoft states that “*some arrangements subject to certain contractual conditions being met*”. In other words, not all of the \$329.1B will convert into current obligations due to contractual conditions not being met. One need only read the news articles to see that datacenters are being delayed and / or construction halted due to a wide range of issues including an increasing focus at the state and local level on halting local datacenter construction.
2. Microsoft does not define these leases as unconditional or non-cancelable which means that in addition to cancellation due to cause there are likely, at least in some of these agreements the right for Microsoft to cancel for a termination fee.

We therefore believe that not all of these not yet commenced additional leases will convert into expenses which would mean that annual lease expenses would likely grow slower.

WHAT ABOUT MICROSOFT OWNED DATACENTERS?

We have no way, at this time, to estimate how much owned datacenters SQ FT will increase but given the relatively small future construction commitments we would suspect that owned as a percentage of total will continue to decrease at least near term. This would both allow Microsoft to focus on delivering Cloud capacity and also decrease upfront cash payments for purchasing land and building datacenters as the company spend aggressively on adding capacity to meet demand. Long term we would expect them to return to building more of their datacenters.

WHY ARE DOLLAR SPENT GROWING FASTER THAN SQ FT?

When we turn to the dollars spent on leases, which are spent predominantly on datacenters, we find that the dollars spent have increased much faster recently than SQ FT at 33% YoY from FY25 to FY26 ([Exhibit 2](#)). While we don't have an exact answer we believe there are 3 reasons:

1. The cost of mixed use datacenters (capable of supporting any proportion of GPU AI servers and traditional CPU servers) are more expensive than traditional CPU only datacenters due at a minimum to power and cooling requirements.

2. Demand for powered datacenters located close to demand has increased and likely driven up the cost for Microsoft (and the rest of the world) for datacenters.
3. In addition, impacting CAPEX is the rebuilding of existing datacenter to support mixed use which has been ongoing since 2019. We suspect that these costs have increased CAPEX and will continue for a while but given the step-down in CAPEX allocated to land, building and **improvement** is shrinking this headwind likely peaked if not is turning a corner.

While CAPEX has been growing fast and the company has accelerated their commitments to future leases (with the vast majority most likely for datacenters), the underlying story is in fact far better than the high level numbers would lead one to believe. In fact, we would argue that Microsoft has been very thoughtful in the growth of current liabilities and future commitments.

MICROSOFT'S HARDWARE COMMITMENTS

Turning next to the last and in some ways most impactful part of spending to add capacity to support Cloud and AI - hardware spending.

In the FY25 and FY26 10Ks Microsoft details "Purchase commitments primarily relate to datacenters and include open purchase orders and take-or-pay contracts that are not presented as construction commitments". These commitments are related to semiconductors, networking equipment, power systems, cooling equipment, power, and other key components of their datacenters.

If we look at the last two 10Ks we find that:

- For FY26 (as of the end of FY25) Microsoft had \$103.9B. For FY27 and beyond Microsoft's purchase commitments were \$6.0B.
- For FY27 (as of the end of FY26) Microsoft had \$169B. For FY28 and beyond Microsoft's purchase commitments were \$25.0B.

In other words, Microsoft is obligating itself for the hardware, infrastructure equipment, power systems and power itself for the next 12 months and has no meaningful commitments beyond that. This would explain why Microsoft announced in Q3FY26 a substantial increase in CAPEX due to hardware price increases - they had no locked-in long-term agreements for hardware components.

We would expect that the quadrupling of purchase commitments beyond 12 months is to lock-in key components (including power and cooling systems) and while it could be for server hardware we believe it for much more than server components.

This would mean that if AI was the bubble that some argue then Microsoft may be stuck with datacenters but not AI specific hardware, which is the far larger cost of AI. The datacenter space could then be repositioned for Microsoft's Cloud and internal use within a couple of years (canceling/ not renewing neocloud contracts, not renewing existing datacenter leases etc.) or subleasing to others for Cloud usage.

WHAT WOULD MICROSOFT DO IF SUPPLY OUTSTRIPS DEMAND?

One of the key concerns that investors have relating to Microsoft (beyond CAPEX growth, which was partially answered with the Q4 guidance) is what would happen if AI supply suddenly was far greater than supply. Whether this was caused by "AI is a bubble" or a major customer having a problem meeting their non-cancelable contractual commitments, the problem is roughly the same with a global bubble worse of course. We believe that the company has built with flexibility in mind and with datacenters SQ FT capable of being used either for CPU or GPU and the decision made at the last minute they can shift new capacity around. They can also delay owned datacenters from coming on-line, cancel lease commitments (even if small fees are required) or even turn the lights off if needed (which would have a relatively small impact on margins and cash).

Amy Hood, in response to our question on the Q4FY26 earnings call laid out many of the components to the answer:

- "... currently, the situation is obviously that demand exceeds available supply in a sort of relatively extreme moment"

- “a lot of the expense, especially you see it in CAPEX, you've seen our CAPEX really pivot toward what I would call and do call short-lived assets,”
- “if the demand environment changes, you just slow down what is in fact the largest component, right, and the driver of COGS”
- “[leases are] a smaller percentage of the overall cost structure... timing can be changed on much of that, especially on the builds or you can stagger the timing of the build-out of ... some of the GPUs and CPUs that you plan to put in.”

In other words, Microsoft has enormous flexibility in how the time and leverage the long-term assets they buy / build and the hardware that they purchase more on demand than in advance. Adding to their own supply side flexibility is the fact that they have a diverse book of business by geo, by segment, by industry. As Amy said: “when you have the ability to late bind some of the more expensive components in short term, you have a big book of business that's flexible.”

The bottom line is that hardware CAPEX, which is currently the biggest component of the CAPEX and the COST is not a long term contracted expense and Microsoft can be incredibly flexible with this. Then combine this with a reasonable mid-teens growth of long-term lease expenses and Microsoft's strong balance and core Cloud portfolio - we believe Microsoft will not be in an over-leveraged position.

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 14: MSFT Income Statement

\$ Millions & Pro Forma, except otherwise noted

	FY2024	FY2025	FY2026	FY2027E	FY2028E	Q127E	Q227E	Q327E	Q427E
Income Statement (PF)									
Revenue	245,122	281,724	331,839	392,630	466,335	91,475	95,942	98,608	106,605
Cost of revenue	74,114	87,831	106,374	129,704	158,894	30,010	32,149	32,821	34,723
Gross Profit	171,008	193,893	225,465	262,926	307,441	61,464	63,793	65,787	71,882
Research and development	29,510	32,488	35,562	39,037	43,331	8,879	9,354	9,807	10,997
Sales and marketing	24,456	25,654	26,710	28,046	31,411	6,003	6,913	7,155	7,975
General and administrative	7,609	7,223	7,956	8,734	9,782	1,969	2,125	2,124	2,516
Goodwill impairment / Employee severance	-	-	-	-	-	-	-	-	-
Total opex & COGS	135,689	153,196	176,602	205,520	243,418	46,861	50,542	51,906	56,210
Operating income	109,433	128,528	155,237	187,110	222,918	44,614	45,400	46,702	50,395
Other income	(1,646)	(4,901)	4,145	(1,200)	(1,200)	(300)	(300)	(300)	(300)
EBT	107,787	123,627	159,382	185,910	221,718	44,314	45,100	46,402	50,095
Taxes	19,651	21,795	30,596	37,182	42,126	8,863	9,020	9,280	10,019
Net income (PF)	88,136	101,832	128,786	148,728	179,591	35,451	36,080	37,121	40,076
Basic EPS (\$)	11.86	13.70	17.34	20.00	24.07	4.77	4.85	4.99	5.38
Diluted EPS (\$, PF)	11.80	13.64	17.28	19.96	24.01	4.76	4.84	4.98	5.37
Basic share count (M)	7,431	7,433	7,429	7,435	7,463	7,428	7,432	7,436	7,444
Diluted share count (M)	7,468	7,465	7,453	7,453	7,480	7,443	7,452	7,453	7,461
Key ratios (PF)									
Gross margin	69.8%	68.8%	67.9%	67.0%	65.9%	67.2%	66.5%	66.7%	67.4%
R&D as % of sales	12.0%	11.5%	10.7%	9.9%	9.3%	9.7%	9.8%	9.9%	10.3%
S&M as % of sales	10.0%	9.1%	8.0%	7.1%	6.7%	6.6%	7.2%	7.3%	7.5%
G&A as % of sales	3.1%	2.6%	2.4%	2.2%	2.1%	2.2%	2.2%	2.2%	2.4%
Operating margin	44.6%	45.6%	46.8%	47.7%	47.8%	48.8%	47.3%	47.4%	47.3%
Tax rate	18.2%	17.6%	19.2%	20.0%	19.0%	20.0%	20.0%	20.0%	20.0%
Net margin	36.0%	36.1%	38.8%	37.9%	38.5%	38.8%	37.6%	37.6%	37.6%
YoY change									
Revenue	16%	15%	18%	18%	19%	17.8%	18.0%	19.0%	18.4%
Cost of revenue	13%	19%	21%	22%	23%	24.8%	23.8%	22.3%	17.6%
Gross Profit	17%	13%	16%	17%	17%	14.6%	15.4%	17.4%	18.8%
R&D	9%	10%	9%	10%	11%	9.0%	10.0%	10.0%	10.0%
S&M	7%	5%	4%	5%	12%	5.0%	5.0%	5.0%	5.0%
G&A	19%	-5%	10%	10%	12%	9.0%	10.0%	10.0%	10.0%
Operating income	22%	17%	21%	21%	19%	17.5%	18.6%	21.6%	24.1%
Net income	20%	16%	26%	15%	21%	15.0%	16.9%	16.8%	13.6%
EPS - diluted	20%	16%	27%	15%	20%	15.3%	17.0%	16.6%	13.3%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 15: MSFT Cash Flow Statement

\$ Millions & GAAP, except otherwise noted

	FY2024	FY2025	FY2026	FY2027E	FY2028E	Q127E	Q227E	Q327E	Q427E
Cash Flow Statement									
Net income	88,136	101,832	133,749	148,728	179,591	35,451	36,080	37,121	40,076
Adjustments to reconcile net income to net cash from operations:									
Depreciation, amortization, and other	-	-	-	-	-	-	-	-	-
noncash items	22,287	34,153	43,448	52,323	75,006	11,105	12,401	13,716	15,100
Stock-based compensation	10,734	11,974	12,405	13,124	14,558	2,995	3,211	3,263	3,655
Deferred income taxes	(4,738)	(7,056)	14,189	(1,581)	(3,197)	(304)	(362)	(415)	(500)
Deferral of unearned revenue, net	5,348	5,438	9,361	9,651	11,303	(8,233)	(3,838)	(4,930)	26,651
Changes in operating assets and liabilities:	(3,524)	(10,788)	(14,256)	(15,655)	(21,525)	12,042	2,795	(8,342)	(22,151)
Other	305	609	(15,961)	-	-	-	-	-	-
Net cash provided by operating activities	118,548	136,162	182,935	206,590	255,736	53,057	50,288	40,413	62,832
Capex	(44,477)	(64,551)	(115,948)	(188,016)	(201,044)	(45,004)	(45,737)	(47,971)	(49,304)
Acquisitions, net of cash	(69,132)	(5,978)	(1,743)	-	-	-	-	-	-
Purchase of securities, net (sale)	34,371	27,705	36,542	19,523	(13,343)	7,461	3,606	13,581	(5,126)
Securities lending payable	-	-	-	-	-	-	-	-	-
Net cash used in investing activities	(96,970)	(72,599)	(139,500)	(168,493)	(214,387)	(37,542)	(42,131)	(34,390)	(54,430)
Net borrowing	575	(8,962)	(3,000)	-	-	-	-	-	-
Share repurchase	(17,254)	(18,420)	(22,271)	(11,134)	(6,489)	(3,944)	(2,907)	(2,829)	(1,454)
Dividends	(21,771)	(24,082)	(26,445)	(29,094)	(32,122)	(6,760)	(7,440)	(7,443)	(7,451)
Other	693	(235)	(830)	1,958	1,961	502	456	505	496
Net cash used in financing	(37,757)	(51,699)	(52,546)	(38,270)	(36,649)	(10,202)	(9,892)	(9,768)	(8,409)
Effect of exchange rates on cash and cash equivalents	(210)	63	(196)	-	-	-	-	-	-
Net change in cash and cash equivalents	(16,389)	11,927	(9,307)	(174)	4,700	5,312	(1,734)	(3,745)	(7)
Cash and cash equivalents at beginning of period	34,704	18,315	30,242	20,935	20,761	20,935	26,247	24,513	20,768
Cash and cash equivalents at end of period	18,315	30,242	20,935	20,761	25,461	26,247	24,513	20,768	20,761

Source: Company reports, Bernstein estimates and analysis

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	7 Aug 2026	Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
			Closing Price			Cur	2026A	2027E	2028E	2026A	2027E	2028E
MSFT (Microsoft)	O	USD	499.99	660.00	(26.4)%	USD	17.28	19.96	24.01	28.9	25.1	20.8
OLD				647.00								
SPX			7,757.64									

PRICE TARGET CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Microsoft Corp**

We value Microsoft shares based on our estimate for 12-month forward NOPLAT per share in 1 year's time (\$24.2), and apply a P/FE multiple of 27x. We then add back in the net cash per share, discounted at 15% to account for potential tax costs and other "friction" to repatriate all cash (\$7.13 per share), arriving at price target of \$660.

RISKS**Microsoft Corp**

Risks to our price target for MSFT include: a decreased demand for Azure PaaS/IaaS; increased Cloud cybersecurity concerns that impact Cloud demand; lower than forecast sales of Microsoft SaaS offerings; increased competitive pricing pressure on IaaS; increased competition in the Office market; decreased PC sales; increased pirating of the company's software, especially Windows and Office; increased functionality in the virtualization layer that limits the need for Windows server on VMs; decreased monetization of Microsoft's online offerings especially Bing; failure to increase international market share of Bing; lower than forecast sales for Xbox products and services; decreases in Windows sales.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors

on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

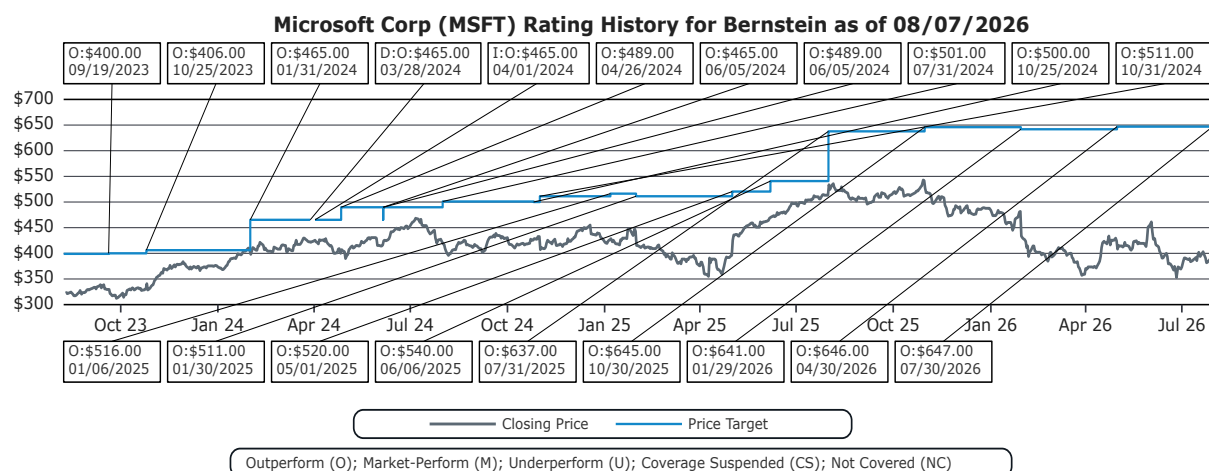
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

Prior to April 1, 2024, Sanford C. Bernstein & Co., LLC. issued the ratings and price target information in the graph(s) below for the following companies: Microsoft Corp.



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

CONFLICTS OF INTEREST

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Microsoft Corp.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Microsoft Corp.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Microsoft Corp.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Microsoft Corp.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Microsoft Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Microsoft Corp.

Shelly Tang has a financial interest in the equity or debt securities of Microsoft Corp (MSFT).

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India

+852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China

+65 Sanford C. Bernstein (Singapore) Private Limited; Singapore

+81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
 - do not have any financial interest in the subject company
 - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
 - is not engaged in any investment banking activities for Indian companies, as such;
 - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
 - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
 - have not received compensation for brokerage services from the subject company in the past twelve months;
 - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and

- do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
- do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment

is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are “eligible counterparties”, “professional clients”, “institutional investors” and/or “professional investors” as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein’s express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein’s express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein’s research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)’ current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients’ home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively “Bloomberg”). Bloomberg or Bloomberg’s licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg’s licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.